Practical no. 19

Aim - Cross-Browser Testing Using Selenium Grid
Execute automation scripts across multiple browsers (Chrome, Firefox, Edge) using Selenium Grid. Compare execution time and compatibility issues.

Tool: Selenium Grid

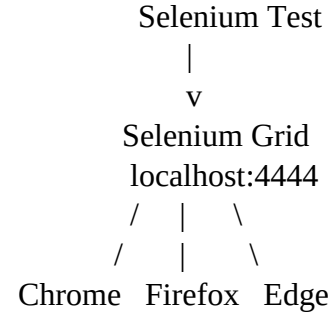
Theory –

1. **Selenium Grid** is a tool used to run Selenium tests on multiple browsers and operating systems.
2. It allows the same automation script to execute on **Chrome, Firefox, and Edge**.
3. Grid uses a **Hub/Node architecture** to distribute test execution across different browser environments.
4. It helps identify **browser compatibility issues** such as differences in page layout, elements, or functionality.
5. Execution time can be recorded and compared for each browser to analyze performance.
6. Cross-browser testing ensures that a web application works correctly and consistently across different browsers.

Step 1 — What you are going to build
You already have:

- Windows 10
- Java
- Eclipse
- Maven
- Selenium 4.35.0
- Chrome
- ChromeDriver
- Your project:
  C:\Users\ITCS\eclipse-workspace\selenium-project

We will create:



The same test will open Google in each browser.

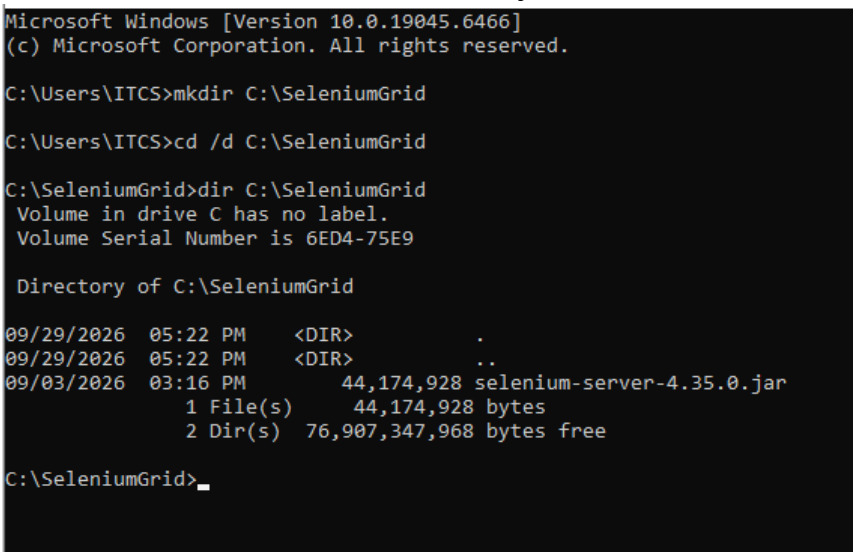
Step 2 — First check Java
Open Command Prompt.
Run**: java -version**
You should see your Java version.

Then: **mvn –version**

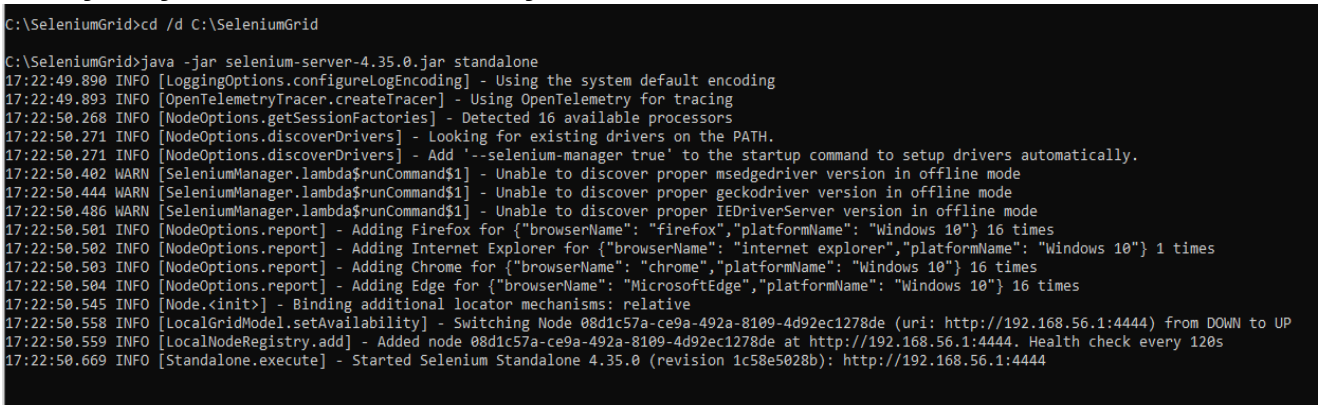
Step 3 — Create a Selenium Grid folder
In Command Prompt and run: **mkdir C:\SeleniumGrid**
Then: **cd /d C:\SeleniumGrid**
Your command prompt should now show:
C:\SeleniumGrid>

Step 4 — Download Selenium Server
You need the Selenium Server JAR.
Download the Selenium Server **4.35.0** JAR because your project is using Selenium 4.35.0.
Put the downloaded file inside: C:\SeleniumGrid
The file should look similar to: selenium-server-4.35.0.jar

Check it
Run: **dir C:\SeleniumGrid**
You should see: selenium-server-4.35.0.jar



Step 5 — Start Selenium Grid
In the same Command Prompt:
cd /d C:\SeleniumGrid
Then: **java -jar selenium-server-4.35.0.jar standalone**



IMPORTANT
Do not close this window.

Leave it running.
Selenium Grid should start on:
http://localhost:4444

Step 6 — Check Selenium Grid in Chrome
Open your normal Chrome browser.
Enter this in the address bar:
http://localhost:4444
Press Enter.
You should see the Selenium Grid page.
If you see the Grid page, **STOP HERE and continue to Step 7**

Step 7 — Open your Eclipse project
Open Eclipse.
From the Eclipse menu:
File → New → Maven Project
Tick on click a simple project
Click **Next**.

Step 8: Enter Project Details
Enter:
Group Id: com.example
Artifact Id: selenium-project
You can leave the other settings as default.
Click: **Finish**

Eclipse will create your project.
You should see something similar to:
selenium-project
├── src/main/java
├── src/test/java
├── JRE System Library
├── Maven Dependencies
└── pom.xml

Step 9: Open pom.xml
In **Project Explorer**, double-click pom.xml
Replace its contents with:
Make sure Selenium is included:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.selenium</groupId>
    <artifactId>selenium-project</artifactId>
    <version>1.0-SNAPSHOT</version>
    <properties>
        <maven.compiler.source>21</maven.compiler.source>
```

```
    <maven.compiler.target>21</maven.compiler.target>
  </properties>
  <dependencies>
    <dependency>
      <groupId>org.seleniumhq.selenium</groupId>
      <artifactId>selenium-java</artifactId>
      <version>4.35.0</version>
    </dependency>
    <dependency>
      <groupId>org.testng</groupId>
      <artifactId>testng</artifactId>
      <version>7.11.0</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

Save – ctrl + S

Step 10 — Create Package
In Eclipse: Right-click:
src/test/java
Select: **New → Package**
Enter: tests
Click **Finish**.

You now have:
src/test/java
└── tests

Step 11 : Create Java Class
Right-click: tests
Select: **New → Class**
Enter: LoginTest
Click **Finish**.
You now have:
src/test/java
└── tests
└── LoginTest.java

Now we will make Chrome work through Grid.
Open: LoginTest.java
For your first Grid test, replace its contents with this:

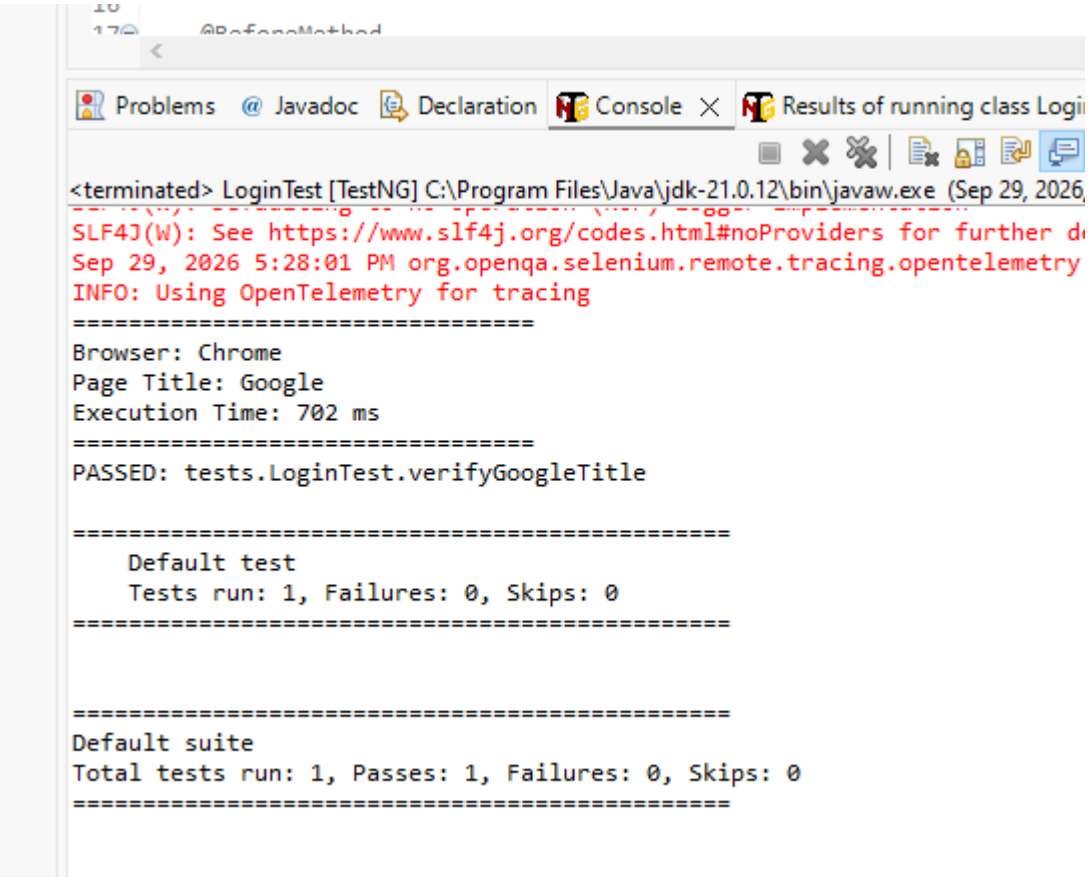
```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
```

```
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
public class LoginTest {
   WebDriver driver;
   @BeforeMethod
   public void setUp() throws Exception {
      ChromeOptions options = new ChromeOptions();
      driver = new RemoteWebDriver(
            new URL("http://localhost:4444"),
            options
      );
   }
   @Test
   public void verifyGoogleTitle() {
      long startTime = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long endTime = System.currentTimeMillis();
      System.out.println("================================");
      System.out.println("Browser: Chrome");
      System.out.println("Page Title: " + title);
      System.out.println("Execution Time: "
            + (endTime - startTime) + " ms");
      System.out.println("================================");
      Assert.assertEquals(title, "Google");
   }
   @AfterMethod
   public void tearDown() {
      if (driver != null) {
         driver.quit();
      }
   }
}
```

Step 12 — Run the Chrome test

In Eclipse: Right-click: LoginTest.java
Choose:
Run As
↓
TestNG Test

Watch the console.
You should see something similar to:



Step 13 — Verify Chrome actually came through Grid
Look at the Command Prompt where you started:
java -jar selenium-server-4.35.0.jar standalone
You should see Grid activity/session information.
You can also refresh:
http://localhost:4444
This confirms that the test is using:
RemoteWebDriver
↓
Selenium Grid
↓
Chrome and **not** directly using ChromeDriver.

Step 14 — Now Firefox
Once Chrome passes, we move to Firefox.
Open LoginTest.java.
Change:

```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
```

```
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
public class LoginTest {
   WebDriver driver;
   @BeforeMethod
   public void setUp() throws Exception {
      FirefoxOptions options = new FirefoxOptions();
      driver = new RemoteWebDriver(
            new URL("http://localhost:4444"),
            options
      );
   }
   @Test
   public void verifyGoogleTitle() {
      long startTime = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long endTime = System.currentTimeMillis();
      System.out.println("================================");
      System.out.println("Browser: Firefox");
      System.out.println("Page Title: " + title);
      System.out.println("Execution Time: "
            + (endTime - startTime) + " ms");
      System.out.println("================================");
      Assert.assertEquals(title, "Google");
   }
   @AfterMethod
   public void tearDown() {
      if (driver != null) {
         driver.quit();
      }
   }
}
```

Step 15 — Run Firefox

Again:

Right-click LoginTest.java

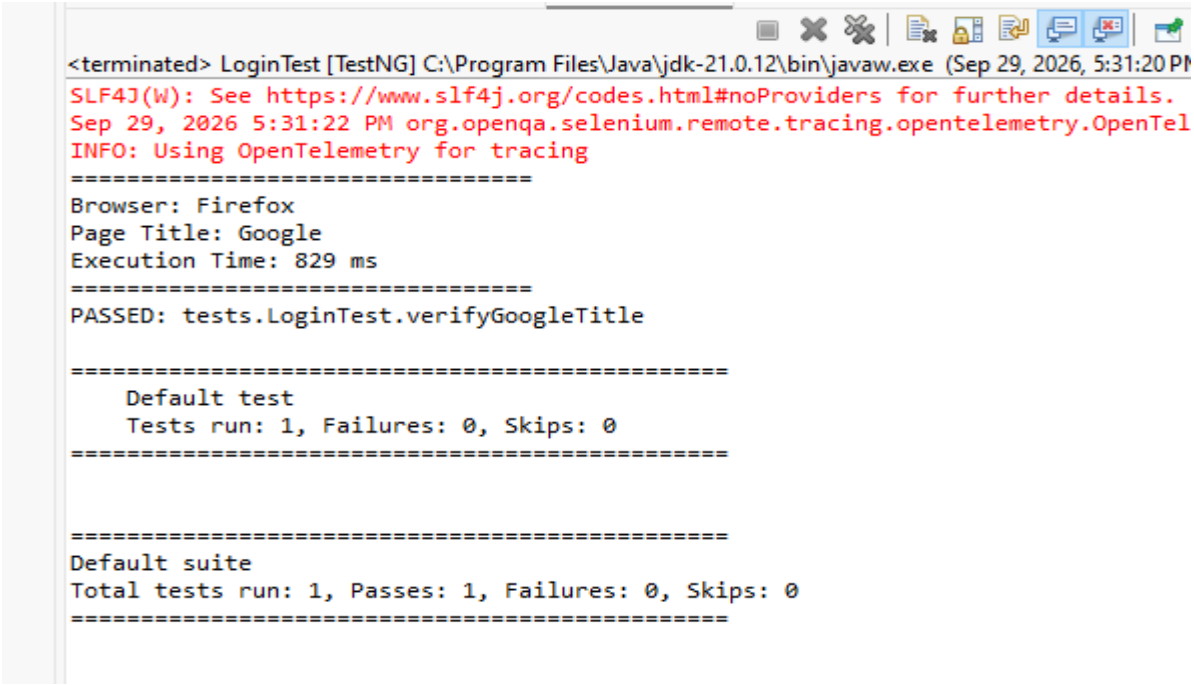
↓

Run As

↓

TestNG Test

Expected:



Step 16 — Now Edge
Just replace the existing code in same java file.

```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.edge.EdgeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
public class LoginTest {
   WebDriver driver;
   @BeforeMethod
   public void setUp() throws Exception {
      EdgeOptions options = new EdgeOptions();
      driver = new RemoteWebDriver(
            new URL("http://localhost:4444"),
            options
      );
   }
   @Test
   public void verifyGoogleTitle() {
      long startTime = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long endTime = System.currentTimeMillis();
      System.out.println("================================");
      EdgeOptions options = new EdgeOptions();
```

```
        System.out.println("Page Title: " + title);
        System.out.println("Execution Time: "
                + (endTime - startTime) + " ms");
        System.out.println("==================================");
        Assert.assertEquals(title, "Google");
    }
    @AfterMethod
    public void tearDown() {
        if (driver != null) {
            driver.quit();
        }
    }
}
```

Step 17 — Run Edge
Again: Right-click LoginTest.java
↓
Run As
↓
TestNG Test

Expected O/P :

```
==================================
Page Title: Google
Execution Time: 438 ms
==================================
PASSED: tests.LoginTest.verifyGoogleTitle

===============================================
    Default test
    Tests run: 1, Failures: 0, Skips: 0
===============================================


===============================================
Default suite
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
===============================================
```

Step 18 — Your final comparison
After running all three browsers, prepare this:

Browser	Page Title	Result	Execution Time
Chrome	Google	PASS	_____
Firefox	Google	PASS	_____
Edge	Google	PASS	_____

Step 19 — Add all three browsers into ONE test
Once you have successfully tested each browser individually, we can make one Java program run all three automatically.

Replace LoginTest.java with:

```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.edge.EdgeOptions;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.Test;
public class LoginTest {
   @Test
   public void chromeTest() throws Exception {
      ChromeOptions options = new ChromeOptions();
      WebDriver driver = new RemoteWebDriver(
            new URL("http://localhost:4444"),
            options
      );
      long start = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long end = System.currentTimeMillis();
      System.out.println("Chrome");
      System.out.println("Title: " + title);
      System.out.println("Time: " + (end - start) + " ms");
      Assert.assertEquals(title, "Google");
      driver.quit();
   }
   @Test
   public void firefoxTest() throws Exception {
      FirefoxOptions options = new FirefoxOptions();
      WebDriver driver = new RemoteWebDriver(
            new URL("http://localhost:4444"),
            options
      );
      long start = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long end = System.currentTimeMillis();
      System.out.println("Firefox");
      System.out.println("Title: " + title);
      System.out.println("Time: " + (end - start) + " ms");
      Assert.assertEquals(title, "Google");
      driver.quit();
   }
```

```
    @Test
    public void edgeTest() throws Exception {
        EdgeOptions options = new EdgeOptions();
        WebDriver driver = new RemoteWebDriver(
                new URL("http://localhost:4444"),
                options
        );
        long start = System.currentTimeMillis();
        driver.get("https://www.google.com");
        String title = driver.getTitle();
        long end = System.currentTimeMillis();
        System.out.println("Edge");
        System.out.println("Title: " + title);
        System.out.println("Time: " + (end - start) + " ms");
        Assert.assertEquals(title, "Google");
        driver.quit();
    }
}
```

Save the code and Run:
Run As → TestNG Test

You should get:

```
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
Sep 29, 2026 5:34:10 PM org.openqa.selenium.remote.tracing.opentelemetry.OpenTelemetryTrac
INFO: Using OpenTelemetry for tracing
Chrome
Title: Google
Time: 705 ms
Edge
Title: Google
Time: 663 ms
Firefox
Title: Google
Time: 1003 ms
PASSED: tests.LoginTest.chromeTest
PASSED: tests.LoginTest.edgeTest
PASSED: tests.LoginTest.firefoxTest

===============================================
    Default test
    Tests run: 3, Failures: 0, Skips: 0
===============================================


===============================================
Default suite
Total tests run: 3, Passes: 3, Failures: 0, Skips: 0
===============================================
```

Step 20 — What if you get an error?
Error: localhost:4444 **cannot be reached**
Go back to your Grid Command Prompt.

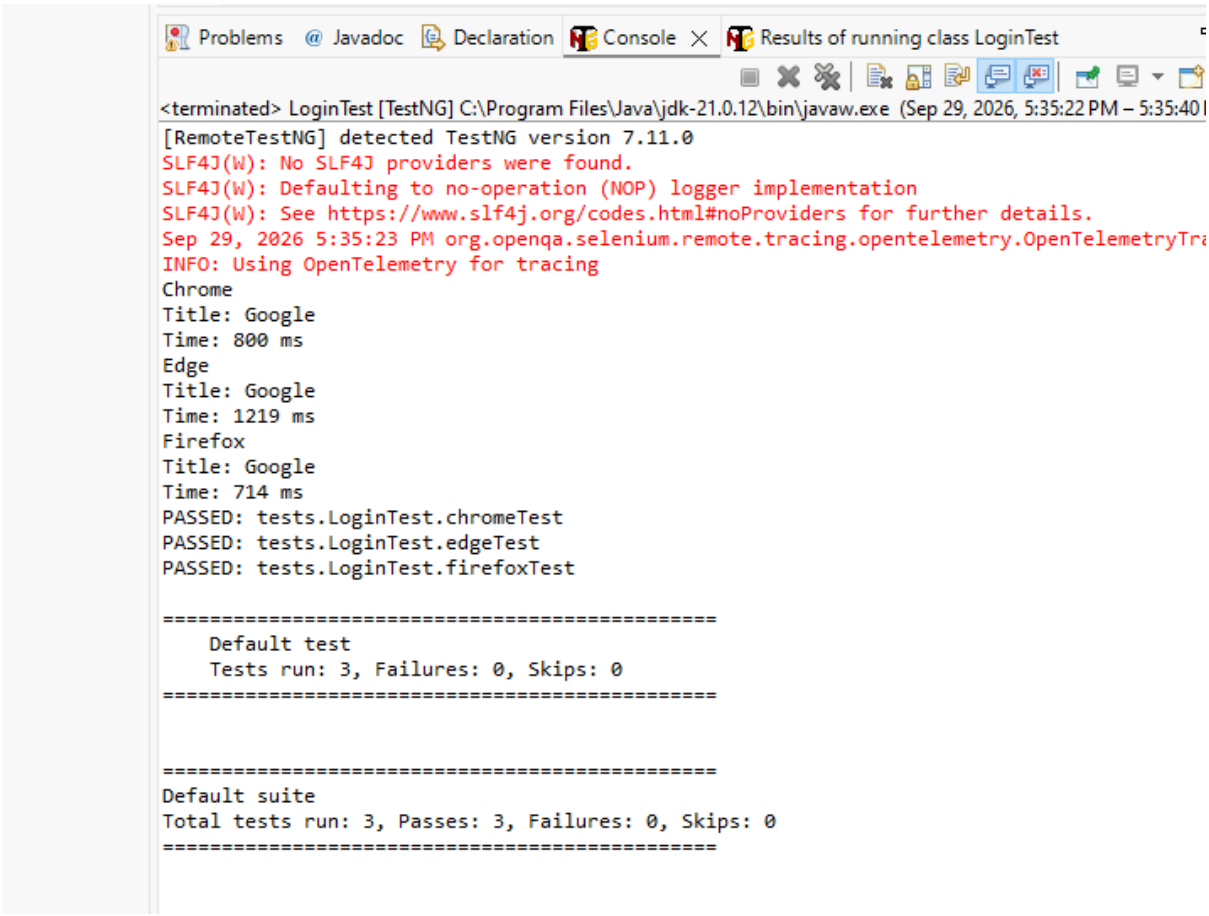
Run:
cd /d C:\SeleniumGrid
Then:
java -jar selenium-server-4.35.0.jar standalone
Keep it open

Step 21 — Working of code –
Very important: don't close Grid
You need two things running:

Window 1 — Selenium Grid
C:\SeleniumGrid>
java -jar selenium-server-4.35.0.jar standalone
Keep this open. **Eclipse — Your test**
Run: LoginTest.java
The flow is:
Eclipse
  |
  | RemoteWebDriver
  ↓
localhost:4444
  |
  ↓
Selenium Grid
  |
  +-------- Chrome
  |
  +-------- Firefox
  |
  +-------- Edge

Final comparison table:

Browser	Result	Execution Time
Chrome	PASS	_____
Firefox	PASS	_____
Edge	PASS	_____


Problems
Javadoc
Declaration
Console
Results of running class LoginTest
<terminated> LoginTest [TestNG] C:\Program Files\Java\jdk-21.0.12\bin\javaw.exe (Sep 29, 2026, 5:35:22 PM – 5:35:40
[RemoteTestNG] detected TestNG version 7.11.0
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
Sep 29, 2026 5:35:23 PM org.openqa.selenium.remote.tracing.opentelemetry.OpenTelemetryTra
INFO: Using OpenTelemetry for tracing
Chrome
Title: Google
Time: 800 ms
Edge
Title: Google
Time: 1219 ms
Firefox
Title: Google
Time: 714 ms
PASSED: tests.LoginTest.chromeTest
PASSED: tests.LoginTest.edgeTest
PASSED: tests.LoginTest.firefoxTest
Default test
Tests run: 3, Failures: 0, Skips: 0
Default suite
Total tests run: 3, Passes: 3, Failures: 0, Skips: 0